



ELK406E

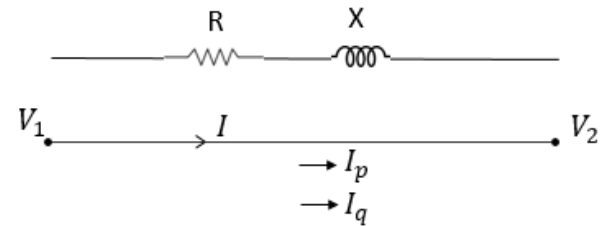
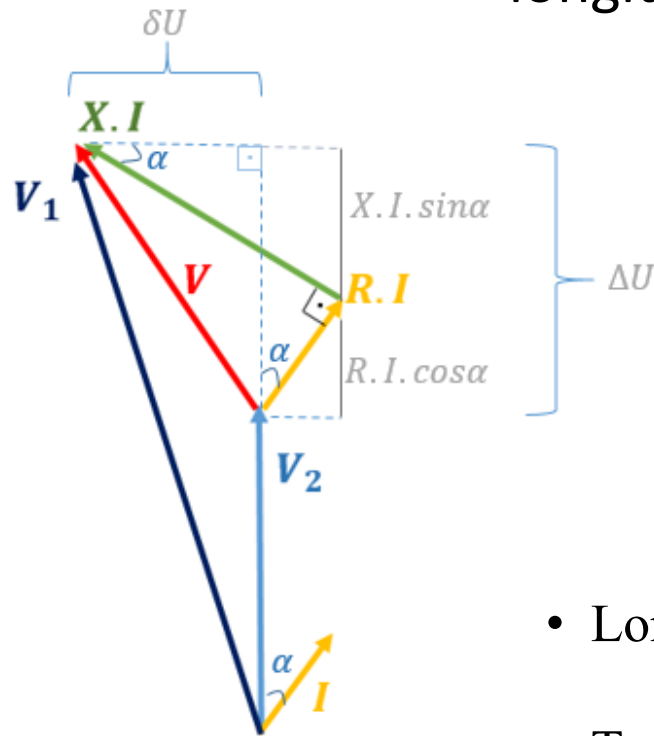
Distribution of Electrical Energy

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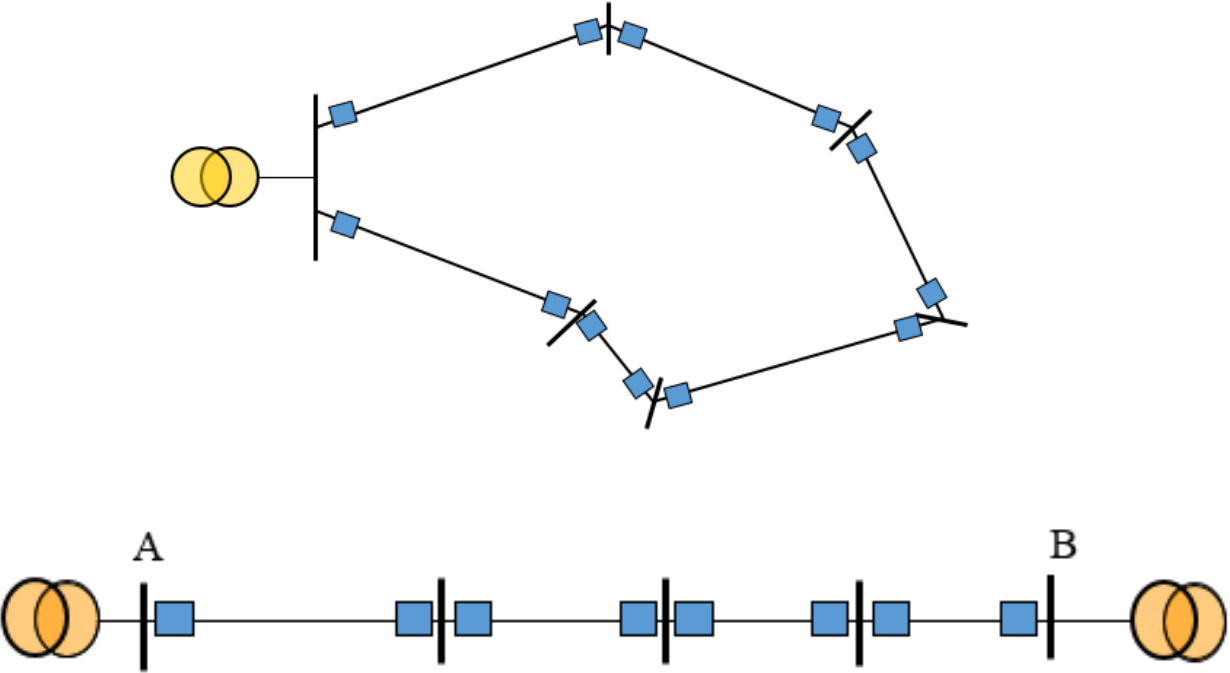
Voltage Drop

- Voltage drop has two components, a longitudinal and a transversal component



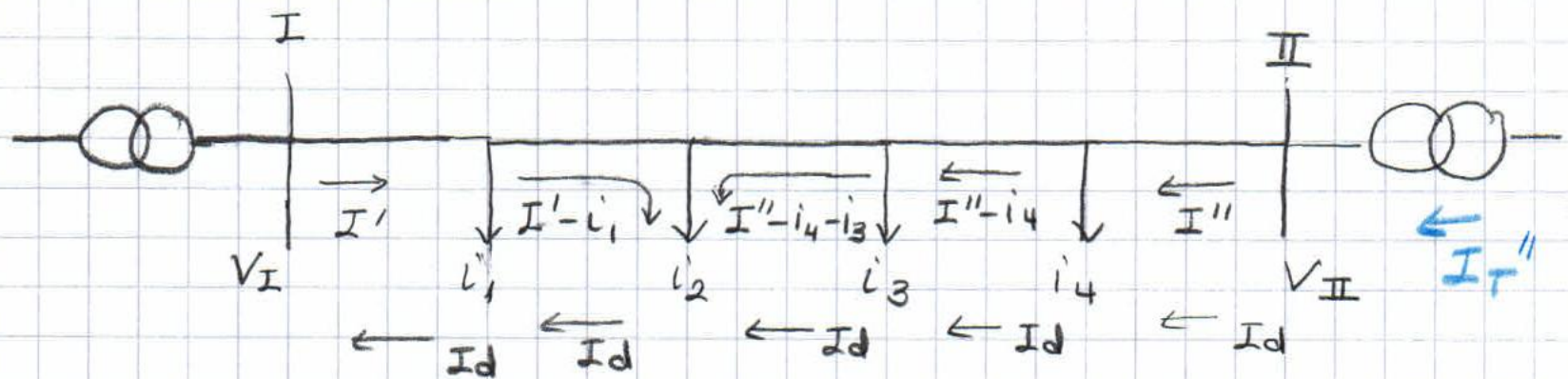
- Longitudinal *Voltage Drop*, $\Delta U = R I_p + X I_q$
- Transversal *Voltage Drop*, $\delta U = X I_p - R I_q$

Closed Loop VD



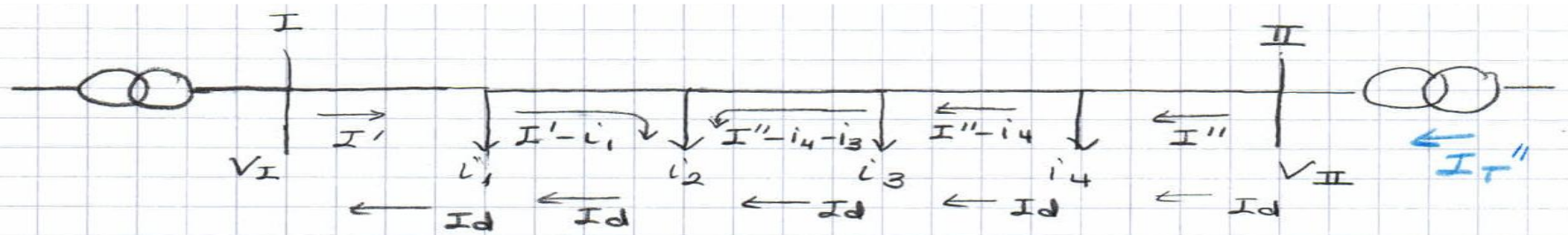
- Feeders being fed at both ends with unequal voltage

Circulating current flows through the feeder from higher potential to lower potential.



If $V_{II} > V_I$, a circulating current I_d flows from II to I.

$$I_d = \frac{V_{II} - V_I}{R} = \frac{V_{II} - V_I}{\left(\frac{L}{XA} \right)}$$



If $V_{II} > V_I$, a circulating current I_d flows from II to I.

$$I'' = \frac{\sum_{j=1}^n \dot{I}_j L_j}{L}$$

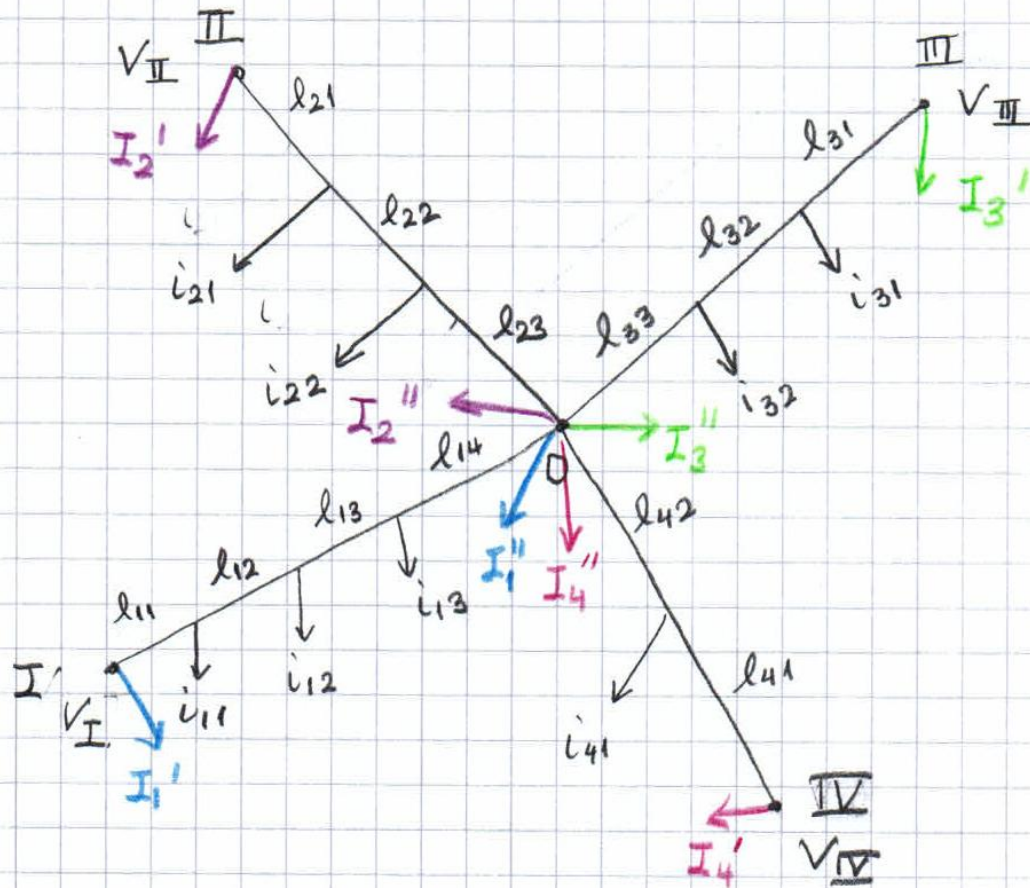
Total current flowing through feeding pt. II.

$$I_T'' = I'' + I_d$$

Find the new critical node where the current changes direction.

— Feeders being fed at multiple ends with unequal voltage

These feeders have a common node. Let's assume that we have four feeders and they're connected to the same common node (D).

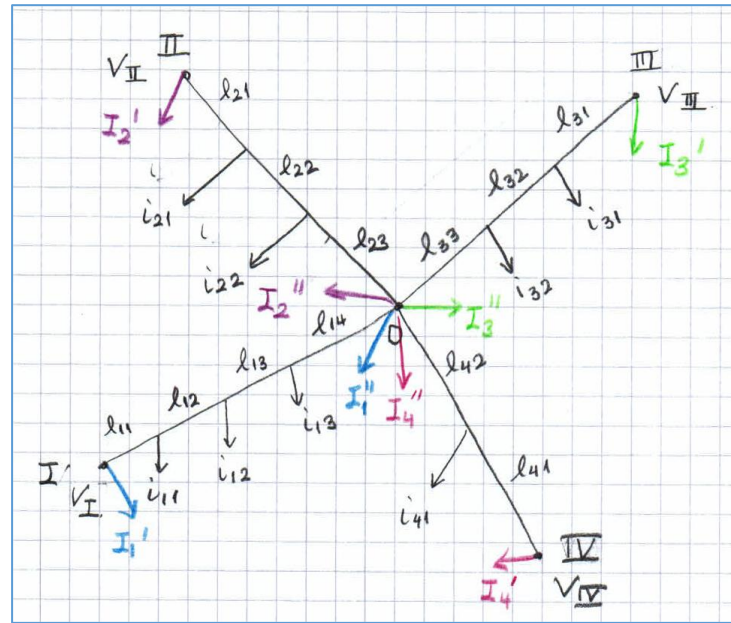


$$I_1'' = \frac{i_{11} l_{11} + i_2 (l_{11} + l_{12}) + i_3 (l_{11} + l_{12} + l_{13})}{l_{11} + l_{12} + l_{13} + l_{14}}$$

$$I_1' = \sum_{j=1}^n i_{1j} - I_1''$$

$$I_2'' = \frac{i_{21} \cdot l_{21} + i_{22} (l_{21} + l_{22})}{l_{21} + l_{22} + l_{23}}$$

$$I_2' = \sum_{j=1}^n i_{2j} - I_2''$$

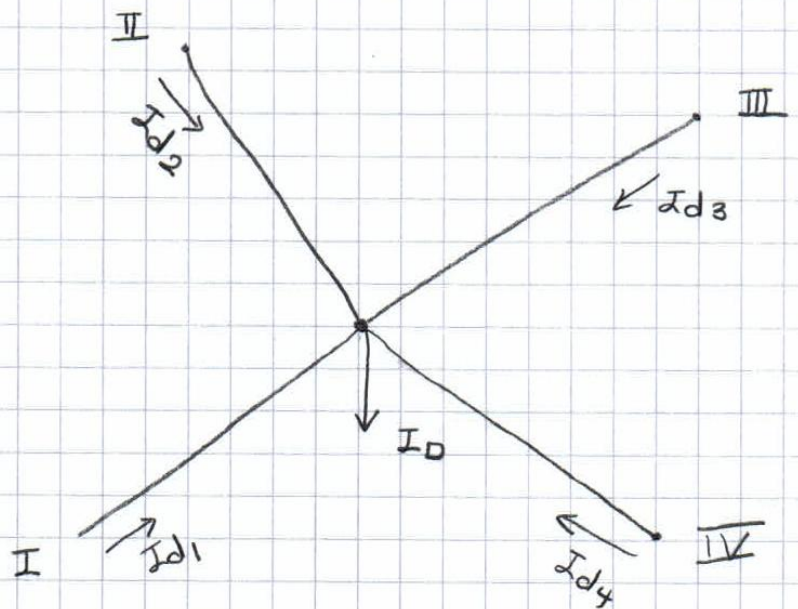


$$I_3'' = \frac{i_{31} l_{31} + i_{32} (l_{31} + l_{32})}{l_{31} + l_{32} + l_{33}}$$

$$I_3' = \sum_{j=1}^n i_{3j} - I_3''$$

$$I_4'' = \frac{i_{41} \cdot l_{41}}{l_{41} + l_{42}}$$

$$I_4' = i_{41} - I_4''$$



I_{di} : circulating current flowing from node i

$$I_D = I_1'' + I_2'' + I_3'' + I_4''$$

$$I_{d1} = \frac{V_I - V_D}{R_1} ; I_{d2} = \frac{V_{II} - V_D}{R_2}$$

$$I_{d3} = \frac{V_{III} - V_D}{R_3} ; I_{d4} = \frac{V_{IV} - V_D}{R_4}$$

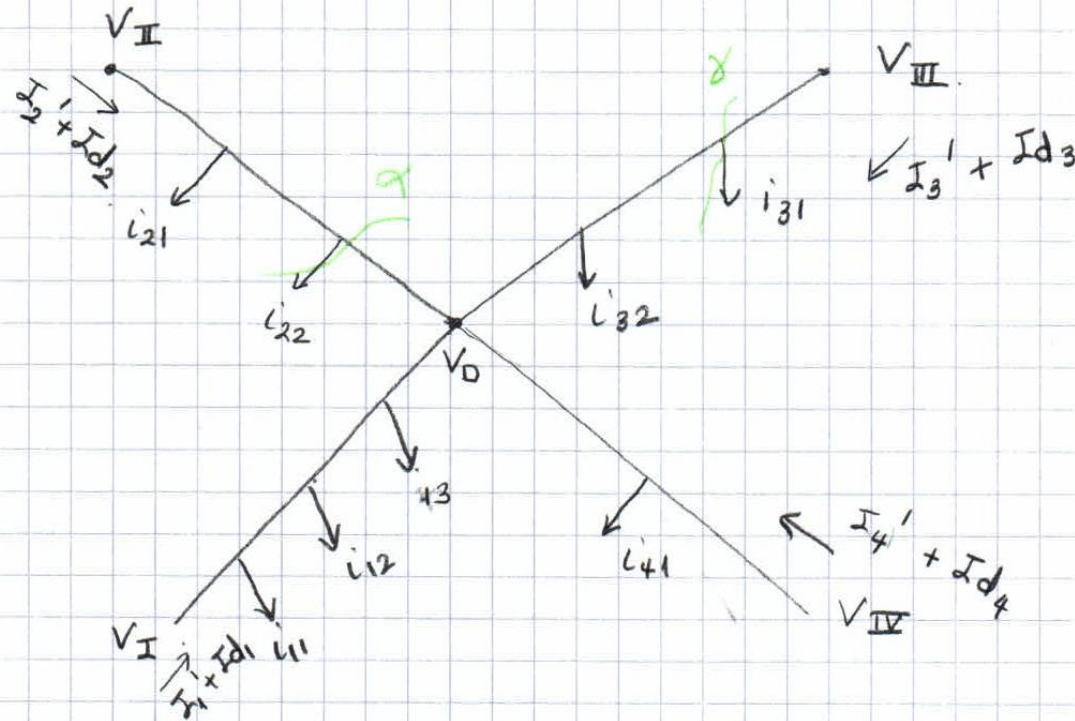
$$I_D = I_{d1} + I_{d2} + I_{d3} + I_{d4}$$

$$I_D = \frac{V_I}{R_1} + \frac{V_{II}}{R_2} + \frac{V_{III}}{R_3} + \frac{V_{IV}}{R_4} - V_D \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$$

$$V_D = R_p \left(\frac{V_I}{R_1} + \frac{V_{II}}{R_2} + \frac{V_{III}}{R_3} + \frac{V_{IV}}{R_4} \right) - I_D \cdot R_p$$

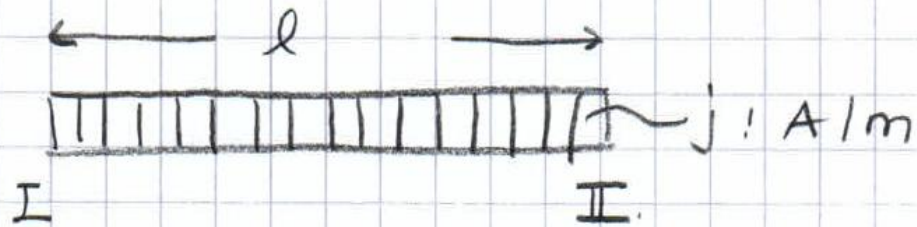
In general assuming that n is the # of nodes (feeding ends)

$$V_D = R_p \left(\sum_{m=1}^n \frac{V_m}{R_m} - I_D \right)$$

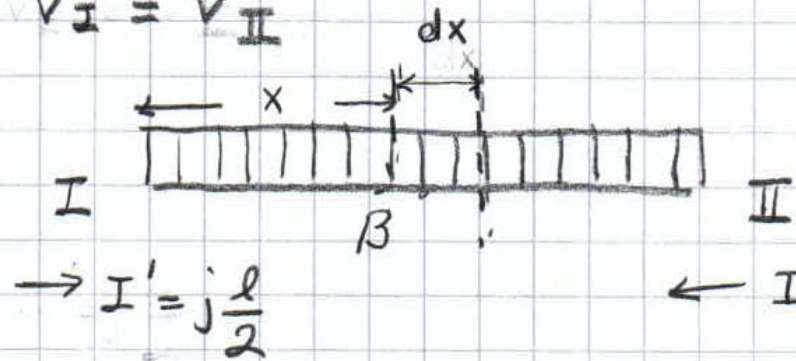


There may be more than one critical node for maximum voltage drop. Calculate voltage drop for each critical node, then find ΔV_{max} and the most critical node.

- Feeders with uniformly distributed load fed at both ends.



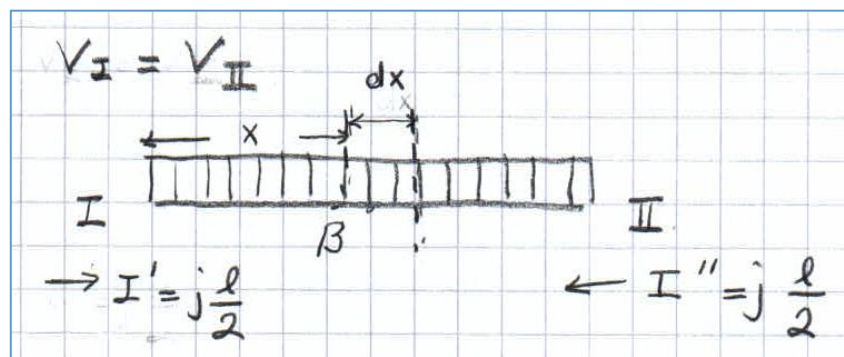
$$V_I = V_{II}$$



The current supplied from each feeding pt. is $j \cdot \frac{l}{2}$

Consider a point B at a distance x meters from I. Then the current at B

$$= j \frac{l}{2} - jx = j \left(\frac{l}{2} - x \right)$$



Now consider a small length dx near pt. B . Its resistance is $r \cdot dx$ and the voltage drop over length dx is

$$dV = j \left(\frac{\ell}{2} - x \right) \cdot r \cdot dx = j \cdot r \cdot \left(\frac{\ell}{2} - x \right) dx$$

$$\Delta V_B = \int_0^x j \cdot r \cdot \left(\frac{\ell}{2} - x \right) dx = j \cdot r \left(\frac{\ell x}{2} - \frac{x^2}{2} \right)$$

$$\Delta V_B = \frac{j \cdot r}{2} (\ell x - x^2)$$

Obviously $\Delta V_{\max}$ occurs at the midpoint .

where $x = \frac{l}{2}$

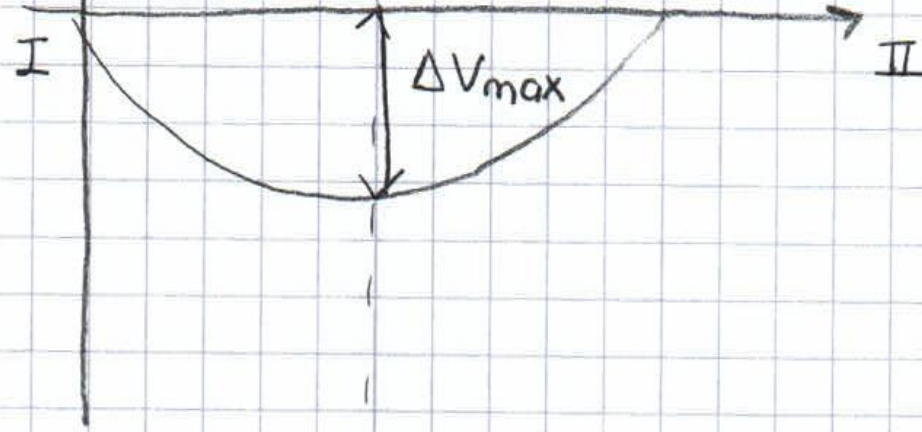
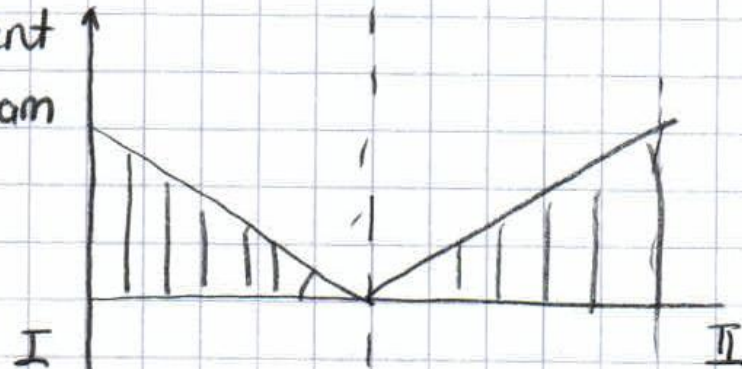
$$\Delta V_{\max} = j \frac{r}{2} \left(\frac{l^2}{2} - \frac{l^2}{4} \right) = j \frac{r}{8} l^2$$

$$\Delta V_{\max} = j \frac{1}{2A} \frac{l^2}{8} \quad \text{or in general} \quad \Delta V_{\max} = j \frac{k}{2A} \frac{l^2}{8}$$

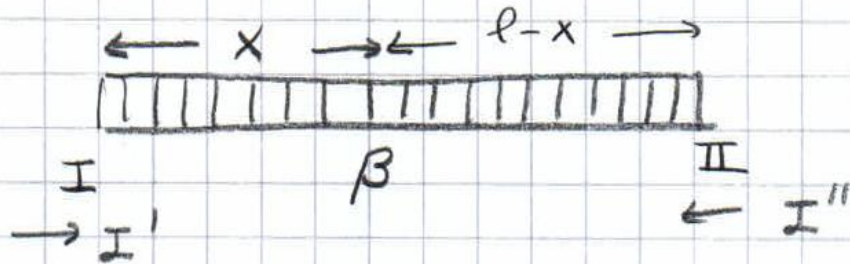


$$V_I = V_{II}$$

current
diagram



$$V_I \neq V_{II}$$

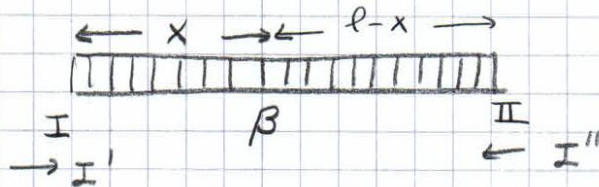


Suppose that the point of minimum potential β is at a distance x meters from I . Then the current supplied by feeding pt. I is $I' = jx$.

$$\Delta V_{I-\beta} = j \cdot r \frac{x^2}{2}$$

$$\Delta V_{II-\beta} = j \cdot r \frac{(l-x)^2}{2}$$

$$V_I \neq V_{II}$$



$$V_B = V_I - \Delta V_{I-B} = V_{II} - \Delta V_{II-B}$$

$$V_I - j \cdot r \cdot \frac{x^2}{2} = V_{II} - j \cdot r \cdot \frac{(l-x)^2}{2}$$

Solving the equation for x , we get

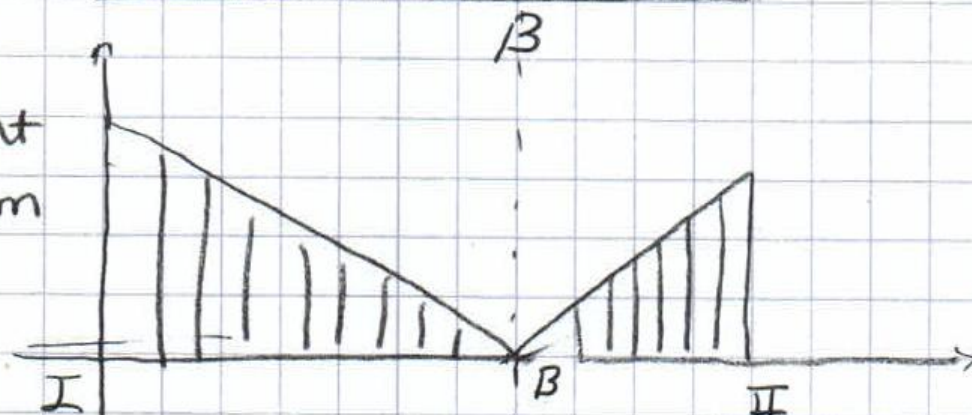
$$x = \frac{V_I - V_{II}}{j \cdot r \cdot l} + \frac{l}{2}$$

If $V_I > V_{II}$ $I_d = \frac{V_I - V_{II}}{r l} = \frac{V_I - V_{II}}{R}$; circulating current

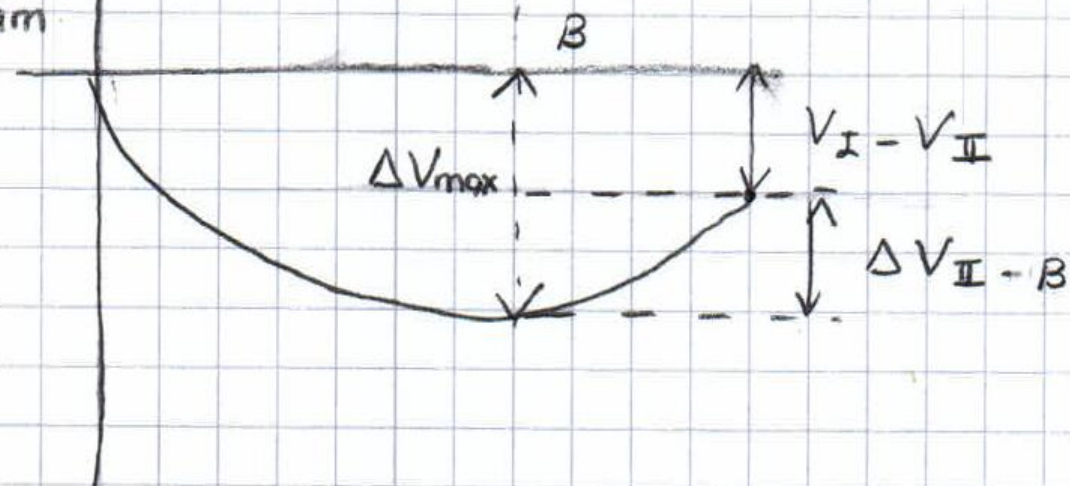
$$x = \frac{I_d}{j} + \frac{l}{2}$$



current
diagram



voltage
diagram



REFERENCES

- T. Gönen “Electric Power Distribution System Engineering”
- Venkata, Lecture Notes
- Electric Power Distribution Handbook